Fourier Orthogonality and Bessel's Inequality

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

A complete authored walkthrough of fourier orthogonality and bessel's inequality, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

HA-01 / Harmonic Analysis / Report v1

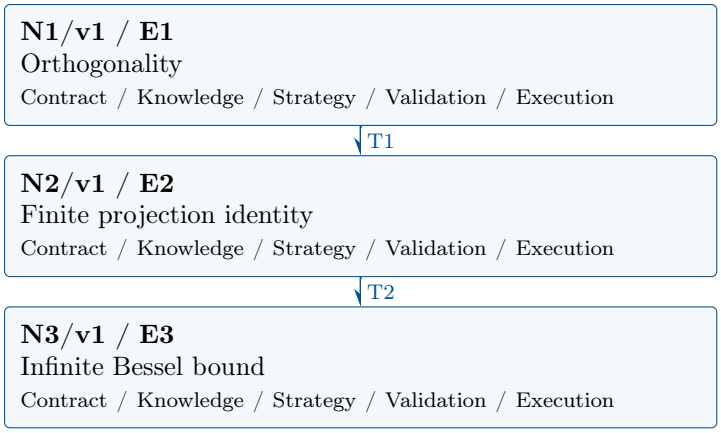
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *On $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ with measure $dx/(2\pi)$, prove e^{inx} form an orthonormal system. For $f \in L^2(\mathbb{T})$ and $c_n = \int f(x)e^{-inx}\,dx/(2\pi)$, prove $\sum_{n\in\mathbb{Z}} |c_n|^2 \le \|f\|_2^2$.*

A. Foundations and permitted imports

Use normalized measure $dx/(2\pi)$ on T. The complex inner product is $\langle f, g\rangle = \int f\ \mathrm{conj}(g)$, linear in the first argument. L^2 functions on this probability space are L^1 by Cauchy–Schwarz, so coefficients are defined. Only finite orthogonal projection and monotone limits of nonnegative sums are needed.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. $\langle e_n, e_m \rangle = \delta_n m$.

Receiving activity. N2: Finite projection identity

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Finite set $F \subset Z$; $c_n = \langle f, e_n \rangle$; prior exports E1/v1

T2: N2/E2/v1 to N3. Exact finite projection identity and finite Bessel bound.

Receiving activity. N3: Infinite Bessel bound

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Bounds for $F = \{-N, \ldots, N\}$; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Orthogonality

Object and role. Integrate frequency differences.

Inputs. $e_n(x) = e^{inx}$; normalized measure; foundational imports stated in §1

C — Contract

Own E1/v1: $\langle e_n, e_m \rangle = \delta_n m$.

K — Knowledge

Retain $e_n(x) = e^{inx}$; normalized measure; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Integrate frequency differences.

V — Validation

A different measure would change the norms and coefficients.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For $m = n$, $\langle e_n, e_m \rangle = \int 1 \, dx/(2\pi) = 1$.

N1.2 For $m \neq n$, integrating $e^{i(n-m)x}$ $over [-\pi, \pi]$ gives zero since its endpoint values agree.

N1.3 Thus the exponentials are orthonormal.

Export E1/v1. $\langle e_n, e_m \rangle = \delta_n m$.

Acceptance gate. A different measure would change the norms and coefficients. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A different measure would change the norms and coefficients. Recheck dependent nodes N2, N3.

B. N2/v1: Finite projection identity

Object and role. Use an orthogonal residual.

Inputs. Finite set $F \subset Z$; $c_n = \langle f, e_n \rangle$; prior exports E1/v1

C — Contract

Own E2/v1: Exact finite projection identity and finite Bessel bound.

K — Knowledge

Retain Finite set $F \subset Z$; $c_n = \langle f, e_n \rangle$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use an orthogonal residual.

V — Validation

Orthonormality alone does not yet imply completeness.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Let $P_F\ f = \sum_{n \in F} c_n e_n$.

N2.2 Orthogonality gives $\langle f - P_F\ f, e_m \rangle = 0$ for $m \in F$ and $\|P_F\ f\|^2 = \sum_{n \in F} |c_n|^2$.

N2.3 Expand the norm of $f = P_F\ f + (f - P_F\ \mathrm{f})$; the cross term vanishes, so $\|f\|^2 = \sum_{n \in F} |c_n|^2 + \|f - P_F\ f\|^2$.

N2.4 Therefore every finite coefficient sum is at most $\|f\|^2$.

Export E2/v1. Exact finite projection identity and finite Bessel bound.

Acceptance gate. Orthonormality alone does not yet imply completeness. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Orthonormality alone does not yet imply completeness. Recheck dependent nodes N3.

C. N3/v1: Infinite Bessel bound

Object and role. Pass to a monotone nonnegative series.

Inputs. Bounds for $F = \{-N, \dots, N\}$; prior exports E2/v1

C — Contract

Own E3/v1: Bessel's inequality under the fixed normalization.

K — Knowledge

Retain Bounds for $F = \{-N, \dots, N\}$; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Pass to a monotone nonnegative series.

V — Validation

Do not replace the inequality by equality before proving density.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 The sums $\sum_{|n|\le N} |c_n|^2$ increase and are bounded above by $\|f\|^2$.

N3.2 Their limit is the nonnegative bilateral series, proving $\sum_Z |c_n|^2 \le \|f\|^2$.

N3.3 No convergence of Fourier partial sums to f has been assumed; equality requires the density bridge in HA-03.

Export E3/v1. Bessel's inequality under the fixed normalization.

Acceptance gate. Do not replace the inequality by equality before proving density. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Do not replace the inequality by equality before proving density. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Orthogonality

1. For $m = n$, $\langle e_n, e_m \rangle = \int 1 \; dx/(2\pi) = 1$.
2. For $m \neq n$, integrating $e^{i(n-m)x} \; over[-\pi, \pi]$ gives zero since its endpoint values agree.
3. Thus the exponentials are orthonormal.

B. §4.N2 — Finite projection identity

1. Let $P_F \; f = \sum_{n \in F} c_n e_n$.
2. Orthogonality gives $\langle f - P_F \; f, e_m \rangle = 0$ for $m \in F$ and $\|P_F \; f\|^2 = \sum_{n \in F} |c_n|^2$.
3. Expand the norm of $f = P_F \; f + (f - P_F \; \mathrm{f})$; the cross term vanishes, so $\|f\|^2 = \sum_{n \in F} |c_n|^2 + \|f - P_F \; f\|^2$.
4. Therefore every finite coefficient sum is at most $\|f\|^2$.

C. §4.N3 — Infinite Bessel bound

1. The sums $\sum_{|n| \leq N} |c_n|^2$ increase and are bounded above by $\|f\|^2$.
2. Their limit is the nonnegative bilateral series, proving $\sum_Z |c_n|^2 \leq \|f\|^2$.
3. No convergence of Fourier partial sums to f has been assumed; equality requires the density bridge in HA-03.

□

V. WORKED ORCHESTRATION

For $f = 2 + 3cosx$, $c_0 = 2$ and $c_{\pm 1} = 3/2$, with all other coefficients 0. Thus the sum is $4 + 2 \cdot 9/4 = 17/2$. Direct integration $of |f|^2$ under normalized measure gives $17/2$, illustrating the finite orthogonal decomposition.

A. Boundary and hypothesis audit

In a general Hilbert space, an orthonormal system need not span the space: a single vector in R^2 misses its perpendicular direction. Validation blocks a candidate equality export from the Bessel proof; Fourier completeness must be supplied separately.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

In a general Hilbert space, an orthonormal system need not span the space: a single vector in R^2 misses its perpendicular direction. Validation blocks a candidate equality export from the Bessel proof; Fourier completeness must be supplied separately.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.